

Lingam Hemang

+91 6302922329 hemanglingam@gmail.com [LinkedIn](#) [GitHub](#) [Portfolio](#)

Professional Summary

Senior Python Backend Engineer who has **designed and owned production backend services and API architecture** for Apple Inc. over 4 years at Infosys – making independent calls on service boundaries, retry/error-handling strategy, and query design across **Django, Flask, and a distributed Spark/Kafka/AWS platform**. Those decisions delivered **40% faster runtimes, 99.5% uptime, and a 35% drop in error rate** on systems I remain accountable for end-to-end. Strong in **OOP, SOLID principles, and API design**; now extending that ownership to **GenAI-backed services** using LangChain, RAG pipelines, and AWS Bedrock.

Education

SRM Institute of Science and Technology

May 2018 – June 2022

Bachelor of Technology – Electronics and Communication Engineering

CGPA: 9.52/10.0

Technical Skills

Languages: Python, SQL, PySpark

Backend & APIs: Django, Flask, REST API Design & Integration, Server-Sent Events (SSE), OOP, SOLID Principles, Design Patterns, pytest, Microservices

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, Snowflake, AWS S3, AWS Lambda, ETL Pipelines, Data Modelling, Schema Design & Validation, Partitioning & Query Optimisation

GenAI / ML: AWS Bedrock, LangChain, RAG Pipelines, Vector Embeddings, ChromaDB, XGBoost, LightGBM, scikit-learn, Prompt Engineering

DevOps & Tools: Docker, Git, CI/CD, Linux, Bash, Splunk, Agile/Scrum, Jira, Confluence

Experience

Infosys

Hyderabad, India

Specialist Programmer – Senior Python Backend Engineer

July 2022 – Present

Client: Apple Inc. | Python, Django, Flask, Spark, Kafka, Airflow, Snowflake, AWS S3, AWS Lambda, REST APIs

- Designed and own **Python backend services** consuming internal and external **REST APIs** (in-house endpoints, Pharos) – chose distributed processing and SQL query optimisation strategies over naive sequential ingestion, cutting end-to-end runtime **40%**.
- Designed **high-availability, fault-tolerant data pipelines** across **AWS S3** and **Kafka** via **Airflow DAGs**; engineered retry logic and self-healing schema validation from scratch – sustaining **99.5% uptime** and cutting error rate **35%**.
- Identified a recurring integration bottleneck and built **reusable Python framework extensions**, cutting onboarding effort **30%**; architected **semantic layer** materialised views for downstream analytics and designed a **Tableau security audit dashboard**.
- Owned **data consistency** for a production application – designed the validation logic behind database views and the load process into a **semantic search cluster**; managed releases via **CI/CD** pipelines across Agile feature teams.

Projects

Agentic AI Git Diff Analyzer | *Python, Django, AWS Bedrock, AWS Lambda, Ollama, ReportLab* | [Repo](#)

- Designed a **dual-mode SQL schema impact analyzer** with a **Django REST backend** – local (Ollama) and **AWS mode** (Bedrock Claude 3.5 via GitHub Webhook → Lambda → EC2); built **LLM-generated impact reports** as styled PDFs and real-time **SSE** streaming with commit-level deduplication.

Document Intelligence Chatbot – RAG Pipeline | *Python, Django, LangChain, ChromaDB, Ollama, EasyOCR* | [Repo](#)

- Built a **10-endpoint Django REST API** powering an end-to-end **RAG pipeline** (LangChain + ChromaDB + Ollama); designed multi-format ingestion (PDF, DOCX, TXT, Images) via **PyMuPDF + EasyOCR**, persistent multi-turn memory, and live model-switching endpoints.

Retail ML Sales Forecasting API | *Python, Flask, scikit-learn, XGBoost* | [Repo](#) **Infosys Hackathon – 2nd Place**

- Built a **multi-model ML forecasting system** (Random Forest, XGBoost, LightGBM) via **Flask REST API**; designed cyclical sin/cos month encoding for seasonal patterns and cross-join **purchase probability scoring**; shipped an interactive **Dash + Plotly** dashboard.